Lydia Woodall

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EDUCATION:

Cornell University, College of Engineering, Ithaca, NY — Expected: May 2028
Bachelors of Science in Mechanical and Aerospace Engineering
Relevant Courses (current *): MAE 3230 Fluid Mechanics*, MAE 3260 System Dynamics*, MAE 3270 Mechanics of Materials*, MAE 2020 Statics, MAE 2030 Dynamics, MAE 2250 Mechanical Design

PROFESSIONAL EXPERIENCE:

Automotive Engineering Intern: AAA National (June-August 2026)

- Designed and built an automated robotic test fixture to perform repeatable headlight measurement sweeps across vehicle height and pitch variations, producing quantitative glare data for headlight analysis.
- Integrated Arduino-controlled actuation with Python data acquisition, developing handshake logic and a GUI to synchronize robot positioning and measurement collection.
- Presented system architecture, design rationale, and automated testing methodology for headlight glare characterization to engineering stakeholders and AAA automotive leadership team.
- Conducted experiments evaluating child left behind vehicle reminder systems and vehicle cabin heating rates under solar exposure across six tests with varied exterior paint colors and manufacturers.
- Aggregated data on vehicle ownership and operating cost to support national vehicle ownership trend analysis.

York County Parks and Recreation Summer Fun Recreation Leader: (June 2025-August 2025)

- Led activities and managed groups of 30+ children ages 6–8, developing communication & teamwork skills

ACADEMIC EXPERIENCE:

Cornell Electric Vehicles Steering Mechanical Subsystem Lead: (November 2024-Present)

- Collaborated with 70+ engineers to build a hyper-efficient autonomous electric vehicle, coming in 5th on the track and winning two Off-track awards and $4500 from Shell Eco Marathon Americas competition
- Led Steering subsystem consisting of Rack & Pinion, Autosteer, Autobrake, Brakes, Accel Pedal, and steering linkages to interface within .05 inch tolerance both with other subsystems and internally
- Designed and optimized Ackermann steering geometry for a custom electric vehicle, improving Ackermann by 10% and targeting 100% at a 10 m turning radius while achieving a 4.5 m minimum radius through increased rack length; applied DFM/DFA principles to reduce manufacturing and assembly complexity
- Designed, manufactured, and tested an autonomous rack and pinion steering system with electromagnetic clutch control; developed a custom 5:1 gearbox with integrated sensor and clutch mounts to deliver 15 Nm from a 3 Nm motor, and built a 15 Nm load-simulating benchtop for automated steering validation
- Designed and manufactured secure housing for data screen, wifi router, and button controls (including turn signal and windshield wipers) experiencing up to 1.5 g forces and heavy vibration mounted to steering wheel supports

VCU Research Assistant: (January 2024-July 2024)

- Performed fault injection experiment on quadcopter drones in VCU Electrical Engineering Lab
- Contributing author on paper published in AIAA: *Realization of a Pervasive Runtime Monitoring Architecture for In-Time Hazard Detection in UAM Systems*

FIRST Robotics Competition Team 422 CEO: (2020-2024)

- Led team of 82 to be a top 1% performance team from 3,000 active competing teams internationally
- Led team meetings, managed annual budget of $100,000, daily code reviews and edits,practice mechanical and software system repair under time pressure, and organized travel for fifty students to four competitions

FIRST Tech Challenge Team 16537 Team Captain: (2019-2024)

- Co-Founded FTC robotics team in rural hometown, designed, CAD modeled, and built semi autonomous systems
- Organized, ran, and judged annual LOGICoyote CAD Challenge with 241 participants (2020-2023)

SKILLS:

CAD: Solidworks (7 years), Fusion360 (5 years), Onshape (4 years), Inventor (3 years)
Programming : Python (5 years), Object Oriented Java (7 years), Arduino (3 years), C++(1 year)
Machining: Manual Lathe, Manual Mill, 3D printing, laser cutting, Bambu Studio

ACTIVITIES:

Cornell Society of Women Engineers - Member (2025-Present)
Cornell Maker Club - Member (2024-Present)